Prodigy Anesthesia Quick Review

Hypothyroidism/Hyperthyroidism

How is peripheral vascular resistance affected by hypothyroidism?	Peripheral vascular resistance is increased and blood volume is reduced. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What are the respiratory changes that occur with hypothyroidism?	The maximum breathing capacity, diffusion capacity, and ventilatory response to hypoxia and hypercarbia are all decreased. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What are the ECG changes often seen with hypothyroidism?	The ECG may exhibit flattened or inverted T waves, low-voltage P waves and QRS complexes, and sinus bradycardia. They are also more prone to the development of ventricular dysrhythmias. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
Patients with hypothyroidism often have nonpitting edema. What is the cause of this?	Hydrophilic mucopolysaccharides accumulate in the dermis, resulting in thickened skin with immobile, nonpitting edema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
How does hypothyroidism affect cardiac status?	There is a decrease in cardiac output due to reductions in both heart rate and stroke volume. Pericardial effusions are common. In advanced hypothyroidism, myocardial contractility is impaired due to both diastolic and systolic dysfunction and cardiomegaly may result. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
How does hypothyroidism affect mental status?	It produces mental sluggishness. Patients develop slow speech, apathy, and listlessness. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is the other name for hypothyroidism?	Myxedema *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is the most common cause of primary hypothyroidism?	Ablation of the thyroid gland with radioactive iodine or surgical removal of the gland. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is the typical TSH level in a patient with hypothyroidism?	5-15 mU/L *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 413.*

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Hypothyroidism/Hyperthyroidism

What is the normal thyroid stimulating hormone (TSH) level?	The normal range is 0.4-5.0 mU/L *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 410.*
What is Hashimoto's thyroiditis and what population does it affect?	Hashimoto's thyroiditis is an autoimmune disease of the thyroid gland that results in enlargement of the gland. It affects middle-aged females. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What electrolyte abnormality is associated with hypothyroidism?	Hyponatremia and impaired free water excretion are common findings in hypothyroidism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is the best approach for treating hypotension following induction of a patient with hypothyroidism?	In patients with hypothyroidism, the administration of alpha agonists such as phenylephrine could substantially increase the systemic vascular resistance against a heart that has limited capacity to compensate by increasing its contractility. Epinephrine 50 mcg would increase both the SVR and contractility, but a 50 mcg dose may be enough to result in hypertension and arrhythmias. Atropine would serve to increase the heart rate but would not address the issue of blood pressure. The best option for these patients is to administer small (2.5 - 5mg) doses of ephedrine while monitoring filling pressures and the ECG closely. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
Is regional anesthesia contraindicated in patients with hypothyroidism?	Regional anesthesia is not contraindicated if there are no coagulation disorders present. In fact, if coagulation is not an issue, regional anesthesia is preferred over general anesthesia for these patients when the surgery permits. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
How does hypothyroidism affect temperature maintenance under anesthesia?	Patients with hypothyroidism are prone to hypothermia which occurs quickly and is difficult to correct once it does occur. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
What are the potential airway considerations for a patient with hypothyroidism?	They are more prone to airway compromise due to edema of the airway and vocal cords. The presence of a goiter may alter the airway anatomy, making intubation more difficult. They also exhibit a prolonged gastric transit time which makes them more susceptible to regurgitation and aspiration of gastric contents during laryngoscopy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*

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Hypothyroidism/Hyperthyroidism

How do patients with hypothyroidism respond to anesthetic drugs?	There are no studies on the anesthetic requirements of patients with hypothyroidism, but most anecdotal evidence in the literature points to an increased sensitivity to anesthesia drugs. This effects are presumed to be due to decreases in cardiac output, blood volume, hepatic metabolism, and renal excretion. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
Is the cardiomyopathy seen with hypothyroidism reversible?	Yes. Patients who take levothyroxine will exhibit improvement in myocardial function within 2-4 months. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389-390.*
How is hypothyroidism diagnosed?	It is diagnosed by the presence of decreased levels of free T4, total T4, T3, R-T3U, free T4 index, and an elevated TSH level. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
What is myxedema coma?	Myxedema coma is an uncommon and severe form of hypothyroidism in which the patient exhibits altered mentation, unconsciousness, hypothermia, hypoventilation, bradycardia, hypotension, and dilutional hyponatremia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 391.*
Is succinylcholine contraindicated in patients with hypothyroidism?	No. Patients with hypothyroidism are at increased risk for aspiration and succinylcholine is often indicated at induction. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
What is primary hypothyroidism?	Primary hypothyroidism is a decrease in the amount of thyroid hormone produced despite adequate levels of TSH. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is the most common treatment for hypothyroidism?	Levothyroxine sodium which restores T3 and T4 levels to normal. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 390.*
What is the cardinal feature of myxedema coma?	Despite the name, unconsciousness is relatively uncommon in myxedema coma. The cardinal symptom is hypothermia, which may be as low as 80 degrees Fahrenheit. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 391.*
What is the only indication for intravenous levothyroxine?	Myxedema coma is the only true indication for intravenous administration of levothyroxine. It is administered as a 300-500 mcg loading dose followed by 50-200 mcg/day. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 391.*

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Hypothyroidism/Hyperthyroidism

What is a goiter?	A goiter is the hypertrophy of the follicular epithelium of the thyroid gland that occurs in response to a reduction in thyroid gland function. The hypertrophy results in enlargement of the thyroid gland itself. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 391.*
What are the hematologic effects of hypothyroidism?	Patients with hypothyroidism exhibit an increased bleeding tendency. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 152.*
In what population of patients does myxedema most commonly occur?	Myxedema coma most commonly occurs in elderly females with a longstanding history of hypothyroidism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 391.*
What is the incidence of hypothyroidism?	It occurs in about 0.5%-0.8% of the general adult population. 20% of women over the age of 60 have subclinical hypothyroidism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is thyroid storm?	Thyroid storm is a life-threatening exacerbation of hyperthyroidism. It most commonly appears in patients who have undiagnosed or untreated hyperthyroidism and respond adversely to the stress of surgery or other stressful event. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329.*
Is the onset of hypothyroidism in adults usually acute or insidious?	Hypothyroidism in adults usually has a slow, insidious progression. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
At what point during the perioperative period is a patient with hyperthyroidism most likely to experience thyrotoxicosis?	Patients can experience thyrotoxic crisis at any point during an anesthetic, but it is most likely to occur within 6 to 18 hours postoperatively. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 734.*
Is ketamine an appropriate induction drug for patients with hyperthyroidism?	No. Drugs that stimulate the sympathetic nervous system such as ketamine, pancuronium, ephedrine, etc., should be avoided whenever possible in patients with hyperthyroidism. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 388.*

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Hypothyroidism/Hyperthyroidism

What are the symptoms of thyroid storm?	Tachycardia, hyperthermia, dysrhythmias, myocardial ischemia, congestive heart failure, and altered mental status. It can closely resemble light anesthesia, pheochromocytoma, and even malignant hyperthermia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329-1330.*
What drug regimen is typically used in thyrotoxic patients prior to surgery to gain control over their symptoms?	A combination of propranolol and potassium iodide are often used prior to surgery to reduce symptoms of circulating levels of T3 and T4. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329.*
What is the most important goal in the anesthetic management of the thyrotoxic patient?	The most important goal is to render the patient euthyroid prior to surgery. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329.*
How long prior to surgery does it normally take to render a patient euthyroid?	Thyrotoxic patients usually require medical management for 7-14 days prior to surgery to create a euthyroid state. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329.*
What population is most affected by hyperthyroidism?	Hyperthyroidism most commonly occurs in female patients between the ages of 20 and 40. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1328.*
What are the three most common causes of hyperthyroidism?	Graves' disease, toxic multinodular goiter, and toxic adenoma. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387.*
A patient exhibits symptoms of thyroid storm postoperatively. What interventions should you take?	Dehydration should be treated with administered glucose-containing intravenous fluids. Cooling measures should be instituted. Beta-blockers should be titrated to reduce the heart rate less than 90 beats per minute. Steroids and antithyroid medications can be administered to reduce circulating levels of thyroid hormones. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 389.*
What is hyperthyroidism?	Hyperthyroidism is defined as the hyperdynamic function of the thyroid gland, resulting in the excessive secretion of thyroid hormones. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387.*

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Hypothyroidism/Hyperthyroidism

What are the general signs and symptoms of hyperthyroidism?	The symptoms of hyperthyroidism include anxiety, emotional lability, heat intolerance, insomnia, fatigue, tremors, exopthalmos, and weight loss. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387.*
What is the leading cause of hyperthyroidism?	Graves' disease, which appears to be an autoimmune syndrome in which thyroid-stimulating antibodies attach to TSH receptors and stimulate the hyperplasia of thyroid tissue. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387.*
How is hyperthyroidism diagnosed?	The hyperdynamic clinical presentation often provides clues, but is confirmed by elevated T3, free T4, Free T4 index, and R-T3U, an increased radioactive iodine uptake, and a low TSH level. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 382.*
How does hyperthyroidism affect the hematocrit? The platelet count?	Hyperthyroidism produces anemia and thrombocytopenia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1329.*
What are the cardiac effects of hyperthyroidism?	The effects of hyperthyroidism on the heart include tachycardia, atrial dysrhythmias, increased myocardial contractility, and cardiomegaly. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387.*
What is toxic multinodular goiter?	This is an extreme form of thyroid enlargement that results from simple goiter in which the thyroid enlargement is severe enough to produce dysphagia and inspiratory stridor. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 387-388.*
What is the first line of treatment for hyperthyroidism?	The first treatment is the administration of propylthiouracil or methimazole. Both of these drugs disrupt the synthesis of thyroid hormone. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 388.*
How does subtotal thyroidectomy compare to radioactive iodine in efficacy in the treatment of hyperthyroidism?	Subtotal thyroidectomy promptly cures thyrotoxicosis in 95% of patients with less risk for hypothyroidism than the use of radioactive iodine. The mortality rate for thyroidectomy is less than 0.1%. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 388.*

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Hypothyroidism/Hyperthyroidism

What is the role of beta-blockers in the treatment of hyperthyroidism?	Beta-blockers relieve many of the symptoms of hyperthyroidism such as anxiety, sweating, heat intolerance, tremors, and tachycardia, but do not address the underlying thyroid disorder. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 388.*

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